

SEMICON Southeast Asia 2026

SEMI **TECH Z** ▶▶ **mers**

Malaysia International Trade and Exhibition Centre | Kuala Lumpur
Date: 4 - 7 May 2026

Challenge Statements



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Company	Challenge Statement	Mentors
	AI Enabled Real Time Safety & Compliance Detection in the Semiconductor Ecosystem	• Vaishnav Subramanian • Ramesh, Akhilass
	Building a Resilient and Sustainable Future for Semiconductor Manufacturing	• Cao Wei Zhong • Daniel Soon Chik Chean
	Challenge Statement: Thermal Precision by Design	• Tan Jun Ji • Calvin Chai • Vincent Lai
	Packaging Reliability Risk Identification	• Mohamad Zainudeen Bin Moideen • Jothisvaran Kumaramohgan
	The "Invisible" Parasitics Problem	• Yuzman Bin Yusoff (SMD) • Derrick Kho (Melexis)

AI Enabled Real Time Safety & Compliance Detection in the Semiconductor Ecosystem



The semiconductor industry operates across a complex ecosystem of upstream and downstream partners, including equipment suppliers, logistics providers, manufacturing fabs, and outsourced assembly and test facilities. Across these environments, safety and compliance critical activities (e.g. vehicle docking, material handling) occur frequently and often rely on manual checks or post incident review. With high traffic, multiple stakeholders, and varying site conditions, real time visibility into safety compliance remains limited, despite the widespread availability of cameras and sensors. Many safety and operational compliance checks in semiconductor operations are manual and labor intensive, reactive rather than preventive and difficult to enforce consistently across sites and partners. This creates gaps in early detection of unsafe or non compliant conditions, increasing risk to people, assets, and operations.

Design a computer vision-based solution that:

- Uses camera input to detect safety or compliance conditions in real time
- Classifies compliant vs. non compliant scenarios under varying environments
- Triggers timely alerts or flags when violations are detected
- Can be conceptually applied across fab, logistics, supplier, or test environments

Desired Outcome is a proposed AI / computer vision approach that:

- Improves real time detection of safety or compliance issues
- Reduces reliance on manual monitoring
- Can scale across multiple partners in the semiconductor value chain
- Demonstrates practical applicability in an industrial environment

How can computer vision be used to improve real time detection of safety or compliance conditions across the semiconductor industry's complex ecosystem, enabling earlier intervention and safer operations?

Building a Resilient and Sustainable Future for Semiconductor Manufacturing



Semiconductor wafer fabs face mounting sustainability challenges due to heavy water usage and high energy demand, making efficient resource utilization essential as many facilities operate in water-stressed regions. Implementing optimized processes and cleaner energy sources helps reduce waste and environmental burden.

Material sustainability is equally critical, as fabrication relies on hazardous chemicals and fluorinated gases; industry efforts now prioritize safer alternatives, material substitution, and lower-emission options to minimize health risks and ecological impact.

Protecting worker well-being requires robust safety systems, strict chemical-handling controls, and responsible waste-disposal practices to prevent exposure and mitigate contamination throughout the fab's lifecycle.

The project should focus on reducing waste through effective reuse and recycling, improving water and energy efficiency, and identifying safer, sustainable material replacements.

It should also explore innovative technologies that close current process gaps—such as advanced recovery systems, low-impact materials, or improved manufacturing methods—to support long-term environmental performance and operational sustainability

Thermal Precision by Design



Semiconductor manufacturing demands extreme precision, consistency, and repeatability. In many critical thermal processes, the furnace is not simply a heating chamber. It is a process enabler that directly influences film quality, material properties, defectivity, yield, and downstream reliability. Even small variations in temperature distribution or cure behavior can lead to non-uniform process outcomes across wafers, across lots, and across time.

Achieving thermal uniformity is not only a hardware problem, but a system-level engineering challenge involving heat transfer, process conditions, equipment behavior, sensing strategy, data interpretation, and process optimization. Likewise, cure profile optimization is not just about reaching a target temperature. It is about controlling the full thermal history experienced by the wafer or material so that the intended physical and chemical transformation is achieved consistently, efficiently, and safely.

This challenge invites you to think like semiconductor process and equipment engineers. Using your knowledge, analytical skills, creativity, and available resources, you will study how a furnace system, can be better understood and improved to achieve tighter thermal uniformity and a more optimized cure profile. Your task is to bridge engineering theory with practical implementation and propose solutions that are technically sound, data-driven, and realistic for an actual semiconductor manufacturing setting.



Packaging Reliability Risk Identification



Packaging-related reliability failures are often detected only during stress testing or after products are released, leading to high cost and delayed corrective action. Current screening methods are largely reactive and do not use manufacturing process data.

Problem Statement:

How can a data-driven tool predict packaging reliability risk early using manufacturing process data?

Focus Area:

Reliability Engineering / Predictive Analytics

The "Invisible" Parasitics Problem



Derived from: Why "Perfect" Schematics Fail in the Real World.

The primary problem students must resolve is the gap between idealized simulation and physical implementation. In a schematic, a line is a perfect conductor with zero resistance or interference. In reality, as frequencies increase and components shrink, those lines gain physical properties that the student didn't "draw."

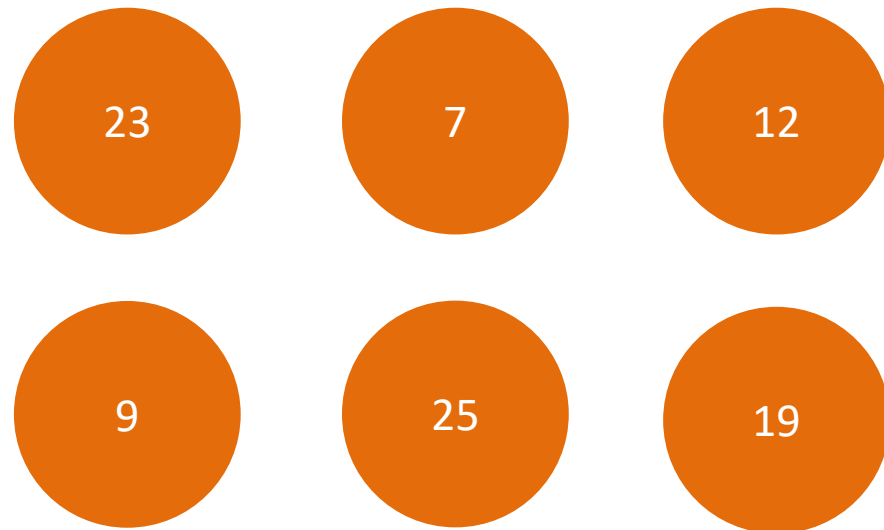
The Student's Task:

Account for Parasitics: How do you compensate for the unintended resistance (R), inductance (L), and capacitance (C) that appear simply because two traces are close together or too long?

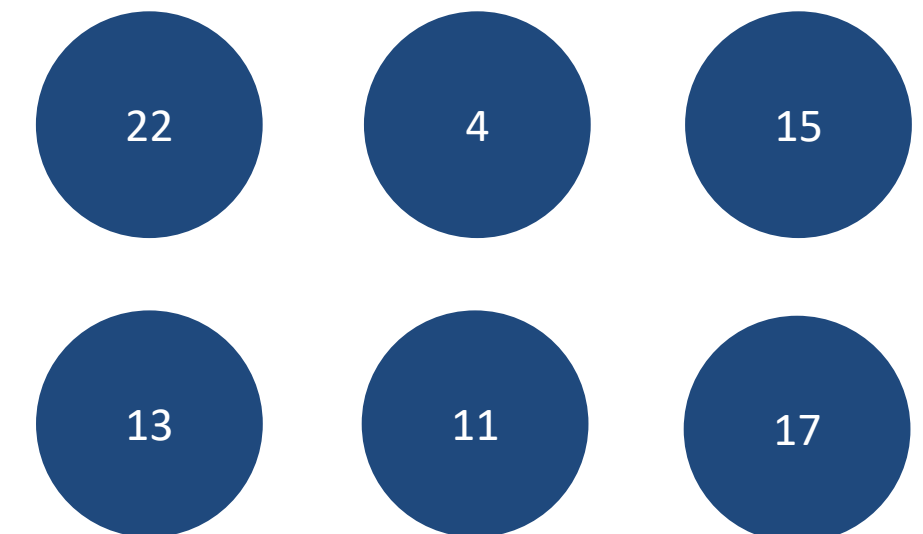
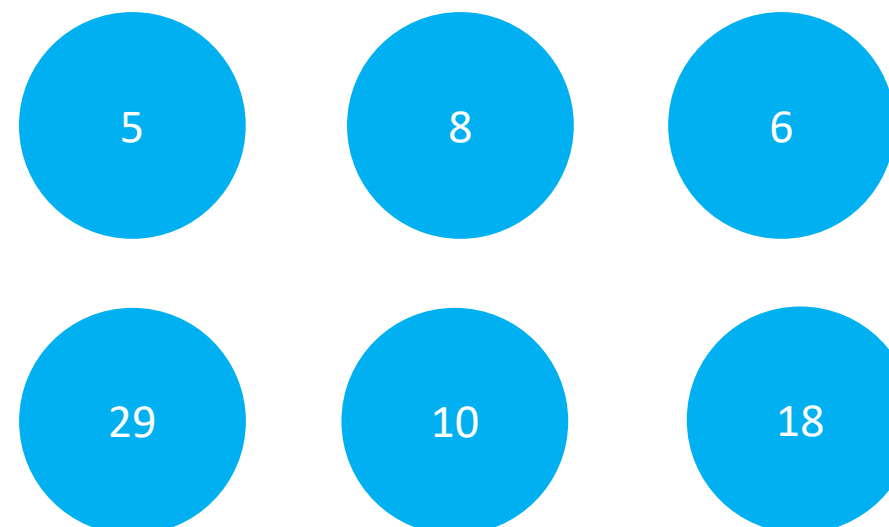
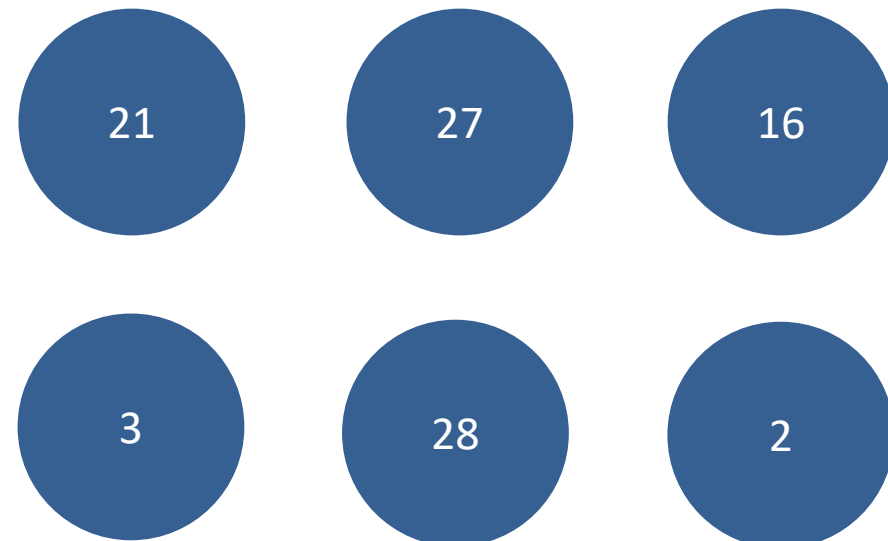
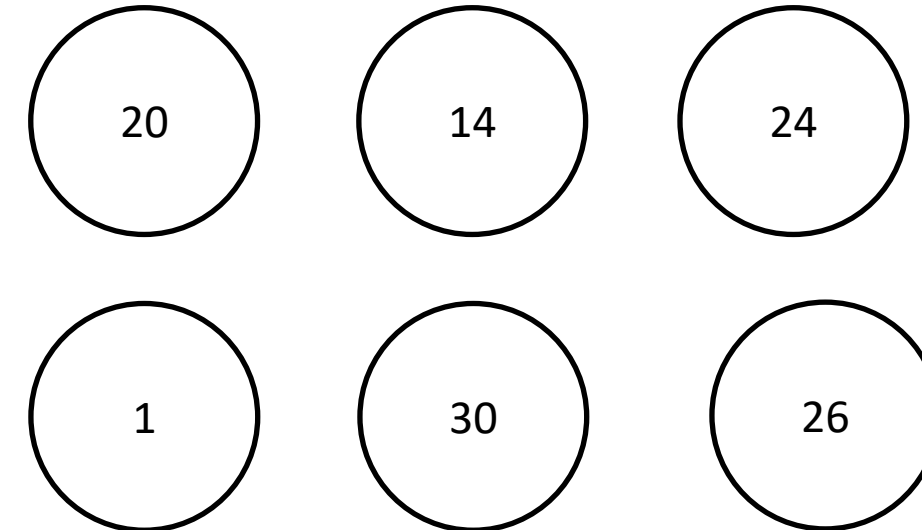
Thermal Management: How do you rearrange a layout so that "hot" components don't cook their "neighbors," even if the schematic says they are connected correctly?

Bridging the Gap: Moving from "does it work in the software?" to "will it survive the fabrication process?"

Challenge Statement



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Presentation



6-May | 14:30 – 15:50h
MITEC
Poster Presentation

- 10-min Poster Presentation (Present your best idea within the time given)
- 5-min Q&A
- Coordinate amongst your teammates

7-May | 10:00 – 12:30h
MITEC | MAINStage
Slides Presentation (Top 5)

- 10-min Slides Presentation
- 10-min Q&A
- Coordinate amongst your teammates
- Save your presentation slides on a USB drive as a backup



Judging Criteria

- Strength of idea
- Commercial Viability
- Teamwork
- Presentation

Bootcamp Prizes

2nd Prize

- MYR 3000 Cash Prize



1st Prize

- MYR 4500 Cash Prize



3rd Prize

- MYR 2000 Cash Prize



Consolation Prize

Gift Set



SEMI Staff Contact Information

Need help? Reach out to us !



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enjoy! the
Event!

